A COUNTEREXAMPLE TO GENERIC INJECTIVITY OF THE TANGENT GAUSS MAP

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We work over $\mathbb{C}$. For an irreducible curve C in an abelian variety A, its *tangent Gauss map* is

$$\phi_C : C^{\mathrm{sm}} \longrightarrow \mathbb{P}(T_0 A), \qquad p \longmapsto d(t_{-p})_p(T_p C),$$

where t_a denotes translation by a. We call C *nondegenerate* if it is not contained in a translate of a proper abelian subvariety. Every automorphism of A has the form $t_a \circ u$, with $u \in \mathrm{Aut}_{\mathrm{gp}}(A)$; its differential is called scalar if $du_0 = \lambda \,\mathrm{id}_{T_0 A}$ for some $\lambda \in \mathbb{C}^\times$.

Theorem 1. *There exist a simple abelian threefold A and a nondegenerate irreducible curve $C \subset A$ whose tangent Gauss map has generic degree 3, but no nonidentity automorphism of A with scalar differential preserves C.*

1. Strategy

1.1. **A pencil whose square gives the Gauss map.** Let X be a smooth nonhyperelliptic curve of genus 4, and let $\mathcal{L}$ be a line bundle of degree 3 with $h^0(X, \mathcal{L}) = 2$. It is base-point-free: otherwise, removing a base point would give a degree-two pencil, making X hyperelliptic. Thus $\mathcal{L}$ defines a degree-three map

$$\psi_{\mathcal{L}} : X \longrightarrow \mathbb{P}^1.$$

The products of its sections define a map to $\mathbb{P}^2$ that factors through $\psi_{\mathcal{L}}$. In a basis $H^0(X, \mathcal{L}) = \langle s_0, s_1 \rangle$, this map is

$$\gamma = [s_0^2 : s_0 s_1 : s_1^2] = \nu_2 \circ \psi_{\mathcal{L}}, \tag{1.1}$$

where $\nu_2 : \mathbb{P}^1 \hookrightarrow \mathbb{P}^2$ is the quadratic Veronese embedding. In particular, γ has degree 3 onto its image.

Our aim is to realize this map as a tangent Gauss map. For this, we seek the following data:

1. a theta characteristic $\mathcal{L}$, that is, $\mathcal{L}^{\otimes 2} \cong \omega_X$, with $h^0(X, \mathcal{L}) = 2$;
2. a quotient $q : J(X) \twoheadrightarrow A$ onto a simple abelian threefold such that

$$q^* H^0(A, \Omega_A^1) = W(X, \mathcal{L}), \qquad W(X, \mathcal{L}) := \mathrm{im}\big(\mathrm{Sym}^2 H^0(X, \mathcal{L}) \longrightarrow H^0(X, \omega_X)\big)\,. \tag{1.2}$$

Here we identify invariant differentials on $J(X)$ with canonical forms on X. We say that $W(X, \mathcal{L})$ *comes from an abelian quotient* when the equality in (1.2) holds, without requiring the quotient to be simple.

For any two independent sections of a line bundle, their three quadratic products are linearly independent: a relation would give a constant quadratic equation for their nonconstant ratio. Thus $W(X, \mathcal{L})$ has dimension 3 whenever $h^0(X, \mathcal{L}) = 2$.

If (1.2) holds, composing an Abel–Jacobi map with q gives $f : X \to A$. The pullbacks of invariant forms are $s_0^2, s_0 s_1, s_1^2$, so its tangent map is precisely (1.1). In Section 3 we show that X is the normalization of $C = f(X)$ and exclude the automorphisms in Theorem 1.

Date: November 3, 2026.

1.2. **Starting from a hyperelliptic curve.** The two requirements above are easier to arrange on a hyperelliptic curve. Recall that a finite map $\pi : (X, p) \to (E, 0)$ to an elliptic curve is *tangential at* p if the elliptic subvariety $\pi^*E \subset J(X)$ is tangent to the Abel–Jacobi image of X at the origin. We identify E with its dual by its canonical principal polarization.

Lemma 1.1. *Let X be a smooth hyperelliptic curve of genus at least 2, and let $p \in X$ be a Weierstrass point. The subspace $H^0(X, \omega_X(-2p))$ comes from an abelian quotient if and only if there is a tangential cover $\pi : (X, p) \to (E, 0)$.*

Proof. At a Weierstrass point,

$$H^0(X, \omega_X(-p)) = H^0(X, \omega_X(-2p)).$$

Indeed, both spaces have dimension $g(X) - 1$, by Riemann–Roch and $h^0(X, \mathcal{O}_X(2p)) = 2$. Under the identification $T_0^* J(X) = H^0(X, \omega_X)$, this space is the annihilator of the tangent line to the Abel–Jacobi image at p.

If π is tangential, take the quotient $q : J(X) \to J(X)/\pi^*E$. Its pulled-back cotangent space is exactly this annihilator. Conversely, suppose a quotient has this pulled-back cotangent space. Its connected kernel K is an elliptic subvariety whose tangent line agrees with that of X at p. Dualizing the inclusion $K \hookrightarrow J(X)$, using the principal polarization of $J(X)$, and composing with the Abel–Jacobi map based at p, gives a nonconstant map $X \to K^\vee$. Its dual elliptic image is K, so the map is tangential at p. □

Lemma 1.2 (The seed)**.** *There exist a smooth hyperelliptic curve X_0 of genus 4, a Weierstrass point $p_0 \in X_0$, and a tangential cover $\pi_0 : (X_0, p_0) \to (E_0, 0)$ such that*

$$A_0 = J(X_0)/\pi_0^* E_0$$

is a simple abelian threefold.

The construction and the arithmetic check of simplicity are given in Appendix A. Fix such a seed and put

$$\mathcal{L}_0 = \mathcal{O}_{X_0}(3p_0), \qquad q_0 : J(X_0) \twoheadrightarrow A_0.$$

The hyperelliptic gap sequence gives $\omega_{X_0} \cong \mathcal{O}_{X_0}(6p_0)$ and $h^0(X_0, \mathcal{L}_0) = 2$. If x is a hyperelliptic coordinate with a double pole at p_0, the sections of $\mathcal{L}_0$ are spanned by $1, x$, and their products by $1, x, x^2$. Therefore

$$\begin{aligned} W_0 := W(X_0, \mathcal{L}_0) &= H^0(X_0, \omega_{X_0}(-2p_0)) \\ &= q_0^* H^0(A_0, \Omega^1_{A_0}). \end{aligned} \tag{1.3}$$

Thus the seed has the required theta characteristic and quotient. We now deform it while preserving these properties and leaving the hyperelliptic locus.

2. A dimension count for deformations

Let B be a sufficiently small, simply connected, marked local deformation space of the spin curve $(X_0, \mathcal{L}_0)$. Write $(X_t, \mathcal{L}_t)$ for the corresponding family. Since theta characteristics deform discretely with the curve,

$$\dim B = h^1(X_0, \mathcal{T}_{X_0}) = 3g - 3 = 9.$$

We impose the two requirements in Section 1 and then estimate the dimension of the hyperelliptic part.

2.1. **The theta-null condition.** Let

$$S = \{t \in B : h^0(X_t, \mathcal{L}_t) \geq 2\}.$$

The spin structure is even. Locally, S is the zero locus of its theta constant, so every component through the seed has dimension at least 8. After shrinking B, upper semicontinuity and parity imply $h^0(X_t, \mathcal{L}_t) = 2$ on S. Cohomology and base change, together with the injectivity of multiplication noted above, then show that

$$W_t = W(X_t, \mathcal{L}_t)$$

forms a holomorphic rank-three subbundle of the Hodge bundle over S. We use the reduced analytic structures on these loci.

2.2. **Preserving the abelian quotient.** Define the rational six-dimensional subspace

$$V_0 = q_0^* H^1(A_0, \mathbb{Q}) \subset H^1(X_0, \mathbb{Q}).$$

Parallel transport under the Gauss–Manin connection gives $V_t \subset H^1(X_t, \mathbb{Q})$. The marking identifies these spaces with a fixed rational vector space. By (1.3),

$$V_{0,\mathbb{C}} = W_0 \oplus \overline{W_0}.$$

For $t \in S$, we have the equivalences

$$\begin{aligned} &\begin{cases} V_t \text{ is a Hodge substructure,} \\ V_{t,\mathbb{C}} \cap H^0(X_t, \omega_{X_t}) = W_t \end{cases} \\ &\qquad \Longleftrightarrow \quad V_{t,\mathbb{C}} = W_t \oplus \overline{W_t} \\ &\qquad \Longleftrightarrow \quad W_t \subset V_{t,\mathbb{C}} \\ &\qquad \Longleftrightarrow \quad \operatorname{rk}\big(W_t \longrightarrow H^1(X_t, \mathbb{C})/V_{t,\mathbb{C}}\big) = 0. \end{aligned} \tag{2.1}$$

For the middle equivalence, rationality makes $V_{t,\mathbb{C}}$ stable under complex conjugation. The two spaces W_t and $\overline{W_t}$ have dimension 3 and intersect trivially by Hodge type, so their direct sum has the full dimension 6 of $V_{t,\mathbb{C}}$.

Let

$$Z = \{t \in S : W_t \subset V_{t,\mathbb{C}}\}.$$

It is the zero locus of a holomorphic bundle map of ranks 3 and 2, hence is locally defined in S by at most six equations. Every irreducible component of Z through the seed therefore satisfies

$$\dim Z \geq 9 - 1 - 6 = 2. \tag{2.2}$$

From now on, fix one such irreducible component and denote it again by Z.

On Z, the space V_t is a polarizable rational Hodge substructure. The correspondence between such weight-one Hodge structures and abelian varieties up to isogeny gives a quotient

$$q_t : J(X_t) \twoheadrightarrow A_t, \qquad q_t^* H^0(A_t, \Omega^1_{A_t}) = W_t.$$

We may take q_t to have connected kernel by choosing the corresponding quotient of $J(X_t)$. Thus W_t continues to come from an abelian quotient.

2.3. **Leaving the hyperelliptic locus.** Write $\mathcal{H}_4 \subset B$ for the hyperelliptic locus. If $t \in S \cap \mathcal{H}_4$, let H_t be the hyperelliptic pencil on X_t. A degree-three line bundle with two sections on such a curve has the form $H_t(p_t)$. Indeed, a base-point-free degree-three pencil, together with the degree-two pencil, would give a birational map to a curve of bidegree $(2, 3)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is 2, contradicting $g(X_t) = 4$. Removing the base point leaves the hyperelliptic pencil.

Now use the theta-characteristic condition. Since $\omega_{X_t} \cong H_t^{\otimes 3}$, the equality $\mathcal{L}_t^{\otimes 2} \cong \omega_{X_t}$ gives $\mathcal{O}_{X_t}(2p_t) \cong H_t$. Thus p_t is a Weierstrass point and

$$\mathcal{L}_t \cong \mathcal{O}_{X_t}(3p_t), \qquad W_t = H^0(X_t, \omega_{X_t}(-2p_t)). \tag{2.3}$$

There are only finitely many choices of the marked Weierstrass point; locally we can work on each marked branch separately.

Lemma 2.1 (Finiteness of tangential covers)**.** *Fix an elliptic curve E and a positive integer n. Up to isomorphism, there are only finitely many smooth hyperelliptic tangential covers $\pi : (X, p) \to (E, 0)$ of degree n.*

Proof. By Treibich's correspondence, these covers are represented by rational irreducible curves in linear systems indexed by $\mu \in \mathbb{Z}^4_{\geq 0}$ and $d \geq 0$, with

$$2n + 1 = \sum_{i=0}^{3} \mu_i^2 + 4d.$$

For fixed n, only finitely many such indices occur. Each corresponding Severi set is finite by [4, Proposition 2.5, 3.]. The correspondence is described in [4, Section 2, Remark 1, 3.–4.]. □

To apply this lemma, we must keep the degree fixed. On Z, let $K_t = (\ker q_t)^0$, an elliptic subvariety of $J(X_t)$. Its integral homology lattice is the annihilator of V_t in $H_1(X_t, \mathbb{Z})$, so it is fixed under the marking. Locally these kernels form a family of elliptic curves. Dualizing their inclusions using the principal polarizations of the Jacobians gives maps

$$\pi_t : (X_t, p_t) \longrightarrow (E_t, 0), \qquad E_t = K_t^\vee.$$

The induced integral cohomology maps are fixed under local markings. Their cup-product pairings therefore show that $\deg \pi_t = n$ is locally constant. By Lemma 1.1 and (2.3), these maps are tangential on the hyperelliptic branch.

Lemma 2.1 gives finite fibers over each elliptic isomorphism class. Working in a local deformation chart introduces only finite choices of spin structure and marked Weierstrass point. Since elliptic curves have a one-dimensional moduli space, it follows that

$$\dim(Z \cap \mathcal{H}_4) \leq 1. \tag{2.4}$$

Together, (2.2) and (2.4) show that Z contains nonhyperelliptic points.

2.4. **Choosing a simple quotient.** It remains to ensure that the quotient at a chosen nonhyperelliptic point is simple. The cup-product polarization restricts to a fixed nondegenerate rational form on V_0. For each proper nonzero rational subspace $R \subset V_0$ on which this form is nondegenerate, let e_R be the orthogonal projector onto R. The locus

$$Z_R = \{t \in Z : e_R(W_t) \subset W_t\}$$

is closed analytic. It is exactly the locus where the transported subspace R is a Hodge substructure. It is proper, since the seed quotient A_0 is simple. Any Hodge substructure is nondegenerate for the polarization, so these are all the subspaces that need to be excluded.

There are only countably many rational subspaces R. By the Baire category theorem, an irreducible complex analytic space cannot be covered by countably many proper closed analytic subsets. We may therefore choose

$$t \in Z \setminus \left(\mathcal{H}_4 \cup \bigcup_R Z_R \right).$$

For this point, $X = X_t$ is nonhyperelliptic and $A = A_t$ is simple. With $\mathcal{L} = \mathcal{L}_t$ and $q = q_t$, we have obtained both requirements of Section 1.

3. Completing the proof

3.1. **Passing to the image curve.** Choose a base point $b \in X$ and set

$$f = q \circ j_b : X \longrightarrow A, \qquad C = f(X),$$

where j_b is the Abel–Jacobi map. The forms $s_0^2, s_0 s_1, s_1^2$ have no common zero, so f is immersive. Moreover, C contains the origin and generates A, because $j_b(X)$ generates $J(X)$ and q is surjective. In particular, C is nondegenerate.

Let Y be the normalization of C, and factor f through a finite map $r : X \to Y$ of degree e. The tangent map γ in (1.1) factors through r, since it is the pullback of the Gauss map on the smooth locus of C. Thus e divides 3. If $e > 1$, then $e = 3$ and the induced map from Y to the conic $\nu_2(\mathbb{P}^1)$ has degree 1. This would make $Y \cong \mathbb{P}^1$, which is impossible: an abelian variety contains no nonconstant rational curve. Hence $e = 1$, and X is the normalization of C. Consequently, ϕ_C has generic degree 3.

3.2. **Excluding scalar symmetries.**

Lemma 3.1. *Let A be a simple complex abelian variety of dimension greater than 1. If $u \in \mathrm{Aut}_{\mathrm{gp}}(A)$ has scalar differential, then $u = [1]$ or $u = [-1]$.*

Proof. Write $du_0 = \lambda \,\mathrm{id}$. On $H_1(A, \mathbb{C})$, the operator u_* is diagonalizable with eigenvalues among $\lambda, \overline{\lambda}$. Since it is rational, the minimal polynomial of λ over $\mathbb{Q}$ has degree at most 2.

If $\lambda \notin \mathbb{R}$, then $K = \mathbb{Q}(\lambda)$ is imaginary quadratic. Write $\lambda = a + ib$, with $b \neq 0$. On $H_1(A, \mathbb{R})$, the complex structure is

$$J = \frac{u_* - a \,\mathrm{id}}{b}.$$

Thus every one-dimensional K-subspace of $H_1(A, \mathbb{Q})$ has a J-invariant realification. Its intersection with the integral lattice defines an elliptic subvariety of A, contradicting simplicity.

Hence $\lambda \in \mathbb{R}$. Taking the trace gives $\lambda \in \mathbb{Q}$. Both u_* and its inverse preserve the integral lattice, so λ is a rational algebraic unit and therefore equals 1 or -1. A homomorphism of complex abelian varieties is determined by its differential, proving the assertion. □

Conclusion of the proof of Theorem 1. Suppose that $\tau = t_a \circ u$ preserves C and has scalar differential. It lifts to an automorphism σ of the normalization X. If σ is the identity, then τ fixes C pointwise. Since $0 \in C$ and C generates A, this forces $a = 0$ and $u = [1]$. We may therefore assume that σ is nontrivial.

A scalar differential preserves every tangent line, so

$$\gamma \circ \sigma = \gamma.$$

Lemma 3.1 gives $u = [1]$ or $u = [-1]$. If $u = [-1]$, then τ is an involution, and its nontrivial lift σ has order 2. The degree-three map γ would factor through $X/\langle\sigma\rangle$, forcing $2 \mid 3$, a contradiction.

If $u = [1]$, then $\tau = t_a$ with $a \neq 0$. Since $g(X) = 4$, the automorphism σ has finite order m. Its cyclic group acts freely on a generic fiber of γ, so $m \mid 3$ and hence $m = 3$. The induced map $X/\langle\sigma\rangle \to \nu_2(\mathbb{P}^1)$ has degree 1, giving $X/\langle\sigma\rangle \cong \mathbb{P}^1$. On the other hand, σ has no fixed point, because $f(\sigma(x)) = f(x) + a$ and a nonzero translation has none. Riemann–Hurwitz for this unramified degree-three quotient would give

$$6 = 2g(X) - 2 = 3\big(2g(\mathbb{P}^1) - 2\big) = -6,$$

again a contradiction. Thus τ is the identity. □

Remark 3.2. The construction gives an immersive normalization $X \to C$. It does not require the image curve C to be smooth or the map $X \to A$ to be an embedding.

Appendix A. A seed with a simple complementary quotient

We construct the seed of Lemma 1.2 from a genus-four Lamé curve. Reduction modulo 11 will verify smoothness and simplicity.

A.1. **The tangential cover.** Set

$$E_0 : y^2 = 4x^3 - 4x - 1,$$

and let e_1, e_2, e_3 be the roots of $4e^3 - 4e - 1$. Define

$$L(B) = (B^3 - 208B + 560)\prod_{i=1}^{3}(B^2 + 10e_iB - 35e_i^2 - 28).$$

This is the genus-four Lamé spectral polynomial for $g_2 = 4$, $g_3 = 1$; see [3, Table 3 and Theorem L]. The associated formulas give a degree-ten cover from the smooth projective model of $\nu^2 = L(B)$ to E_0, sending the point at infinity to the origin. The cover is tangential at infinity: the Lamé operator has an odd-order commuting operator with elliptic coefficients, and its spectral cover is tangential by [2, Proposition 3.1 and the paragraph after equation (17)].

The polynomial and cover are defined over $\mathbb{Q}$. The change of variable $Y = 4\nu$ gives the integral model $Y^2 = 16L(B)$. The Weierstrass discriminant of E_0 is 37, and reduction of L modulo 11 gives

$$\overline{L} = B^9 + B^4 + 6B^3 + 9B^2 + 7B + 5$$

with factorization

$$\overline{L} = (B + 2)(B^2 + 9B + 5)(B^3 + 5B^2 + 6B + 1)(B^3 + 6B^2 + 7B + 6).$$

These are distinct irreducible factors over $\mathbb{F}_{11}$. Thus the curve X_0 is smooth of genus 4 and has good reduction at 11. Its unique point at infinity, denoted p_0, is a Weierstrass point. This supplies the required tangential cover π_0.

A.2. **Checking simplicity.** Let $N_r = \#X_0(\mathbb{F}_{11^r})$. Direct point counting gives

$$(N_1, N_2, N_3, N_4) = (10, 120, 1312, 15312).$$

These numbers can be reproduced from

$$N_r = 11^r + 1 + \sum_{b\in\mathbb{F}_{11^r}} \chi_r(\overline{L}(b)),$$

where χ_r is the quadratic character extended by $\chi_r(0) = 0$. For $r = 2, 3, 4$, one may use the respective irreducible defining polynomials

$$z^2 + 1, \qquad z^3 + 4z^2 + 1, \qquad z^4 + 4z^3 + 1.$$

The four character sums are $-2, -2, -20, 670$. Newton's identities and the functional equation give the Frobenius polynomial

$$\begin{aligned} P_{X_0,11}(T) = {} & T^8 - 2T^7 + T^6 - 6T^5 + 180T^4 \\ & - 66T^3 + 121T^2 - 2662T + 14641. \end{aligned}$$

Also, $\#E_0(\mathbb{F}_{11}) = 7$, so $P_{E_0,11}(T) = T^2 - 5T + 11$.

The maps on Jacobians extend across good reduction. Since $\pi_{0,*}\pi_0^* = [10]$ and $11 \nmid 10$, the elliptic factor remains an elliptic factor in the special fiber. The quotient $A_0 = J(X_0)/\pi_0^*E_0$ has good reduction, with Frobenius polynomial

$$\begin{aligned} h(T) &= P_{X_0,11}(T)/P_{E_0,11}(T) \\ &= T^6 + 3T^5 + 5T^4 - 14T^3 + 55T^2 + 363T + 1331. \end{aligned}$$

Modulo 3, this becomes $\overline{h} = T^6 - T^4 + T^3 + T^2 - 1$. The finite-field irreducibility test gives

$$\gcd(\overline{h}, T^9 - T) = \gcd(\overline{h}, T^{27} - T) = 1, \qquad T^{729} \equiv T \pmod{\overline{h}}.$$

Hence h is irreducible over $\mathbb{Q}$, and the special fiber of A_0 is simple. It is ordinary because $11 \nmid 14$. By [1, Proposition 3.4], an ordinary simple abelian threefold over a finite field of odd cardinality is absolutely simple unless its Frobenius polynomial has the form $T^6 + cT^3 + q^3$. Our polynomial does not have this form, so the special fiber is absolutely simple.

Absolute simplicity then holds in characteristic zero: a proper positive-dimensional geometric abelian subvariety would, after a finite base extension, specialize to such a subvariety in the good special fiber. Thus A_0 is simple over $\mathbb{C}$, proving Lemma 1.2.

The accompanying script `verify_tangent_seed.py` reproduces the polynomial reduction, point counts, Frobenius factorization, and irreducibility checks using only the Python standard library.

References

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